Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Perchonok, Michele Ph.D.		
Project Title:	Themostabilized Food Development		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The National Aeronautics and Space Administration (NASA) is working towards future long duration manned space flights beyond low earth orbit. The duration of these missions may be as long as 2.5 years and will likely include a stay on a lunar or planetary surface. The transit portion of the food system will include a stored food system similar to that used on the International Space Station (ISS). The major technological hurdle for this stored food system is the shelf-life extension of the products to 3 - 5 years, while maintaining acceptability, nutrient content, and minimal packaging. With the need for longer shelf life, the menu composition will move more towards thermostabilized foods and away from freeze-dried foods. The objective of this research is to develop two new thermostabilized food items for ISS and long duration missions. The foods will provide more variety in the ISS menu especially in the areas that are lacking currently such as breakfast items and vegetable side dishes. The development of more thermostabilized foods would not only augment the ISS food		

	system but would also provide the variety, nutrition, and acceptability that will be needed for the future long duration space flights beyond low earth orbit. A standard food product development process will be utilized to develop the food items. Sensory and analytical testing will be conducted on the developed items to insure that the foods are safe and acceptable. To develop new products that will have a minimum of a 3 year shelf life, be nutritious, and have an overall acceptablility of at least 6.0 when measured on a 9-point hedonic scale. As missions become longer, the crew will require more variety in their menu. These new items will provide the variety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These pouched, shelf stable products may be used for camping trips and for meals during electrical black outs.
Task Progress:	Two new products were developed this fiscal year, broccoli souffle and vegetable omelet.
Bibliography Type:	Description: (Last Updated: 01/30/2012)